

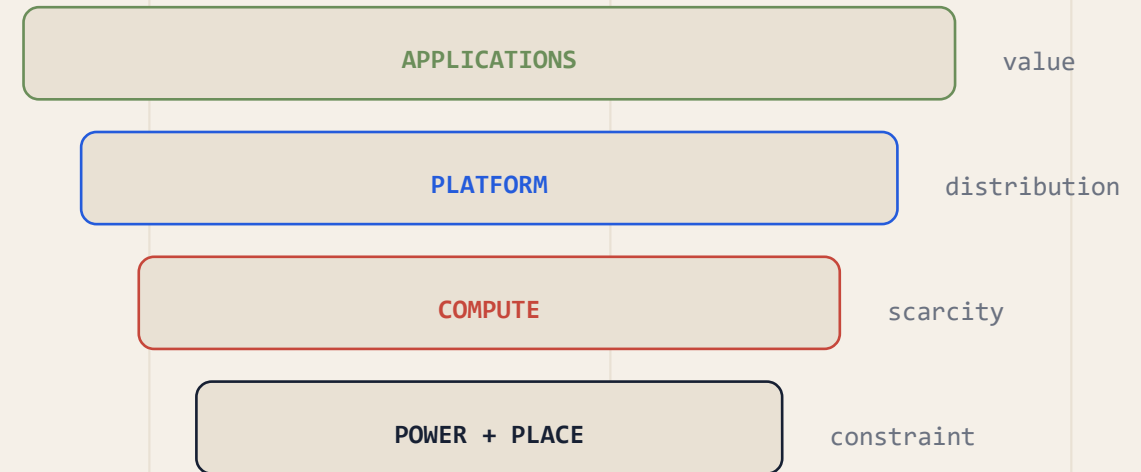
AI INFRASTRUCTURE ECONOMICS

Capital moves first. Constraints decide what ships.

A report on capacity, capital, and the physical stack.

A source-led editorial data study
for strategy and platform operators.

THE VALUE CHAIN IS PHYSICAL



PHYSICAL BEFORE DIGITAL

AI is not one market. It is a stack of dependencies.

The margin pool can move upward; the bottleneck often stays physical.

07 / APPLICATIONS

workflow · distribution · willingness to pay

06 / MODELS

training · inference · evaluation

05 / SOFTWARE

orchestration · serving · observability

04 / SYSTEMS

accelerator · memory · network fabric

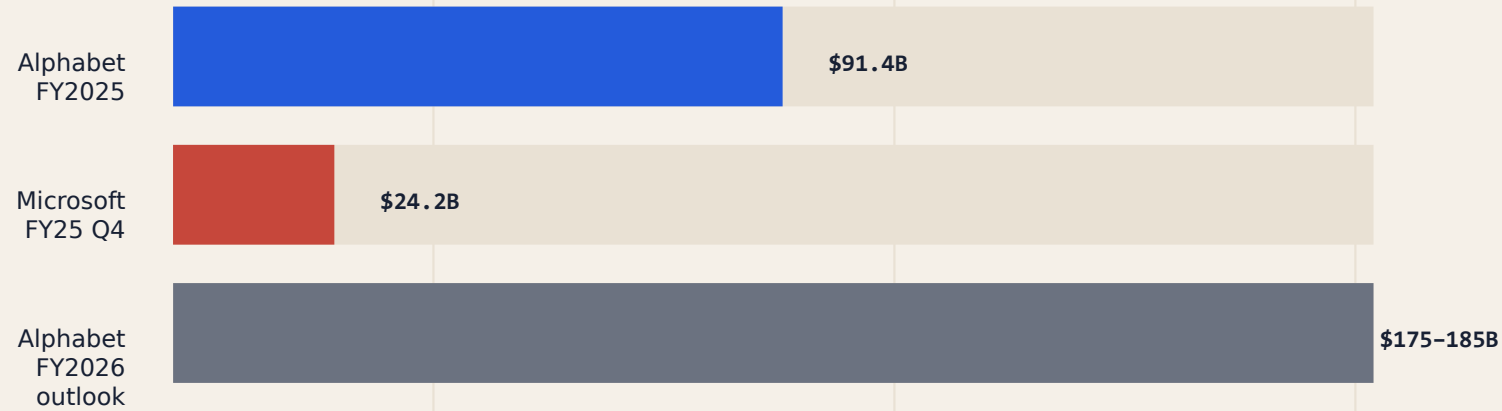
03 / DATA CENTERS

land · cooling · power delivery

dependency ↑

Capital is accelerating — but the periods are not interchangeable.

Reported figures are shown with company and fiscal-period labels, not as a false league table.



READ THIS AS

directional evidence
not a single-period ranking
different disclosure periods
different business mix

WHAT IS REPORTED

Large technical-infrastructure commitments are visible.

WHAT IT SUGGESTS

Capacity is becoming a strategic balance-sheet item.

WHAT IT DOES NOT PROVE

CapEx alone does not tell us who captures the margin.

Where the money lands changes the depreciation story.

One disclosed split makes the physical stack visible.

ALPHABET FY2025

\$91.4B

technical infrastructure CapEx

**The stack has
different clocks.**

A server cycle can move faster
than a building or grid connection.

APPROXIMATE 2025 COMPOSITION

60% SERVERS

40% DC + NETWORK

SHORTER CYCLE

servers
faster refresh
utilization pressure

LONGER DURATION

data centers
networking
depreciation horizon

The constraint is not “compute.” It is the coordination of four physical systems.

Land, energy, networking, and servers arrive on different timelines.

ARRIVAL ORDER

01

02

03

04

LAND

permitted + buildable
lead time $\neq$ demand

ENERGY

predictable supply
lead time $\neq$ demand

NETWORK

fabric + interconnect
lead time $\neq$ demand

SERVERS

GPU + components
lead time $\neq$ demand

CAPACITY ONLY SHIPS WHEN THE SLOWEST LAYER ARRIVES.

This is why supply-chain risk belongs inside the AI strategy — not in a footnote after the model roadmap.

Capacity becomes economics only through utilization.

The flywheel is simple to draw — and difficult to keep turning.

CAPACITY

hardware + power

UTILIZATION

load + scheduling

REVENUE

serving demand

CASH

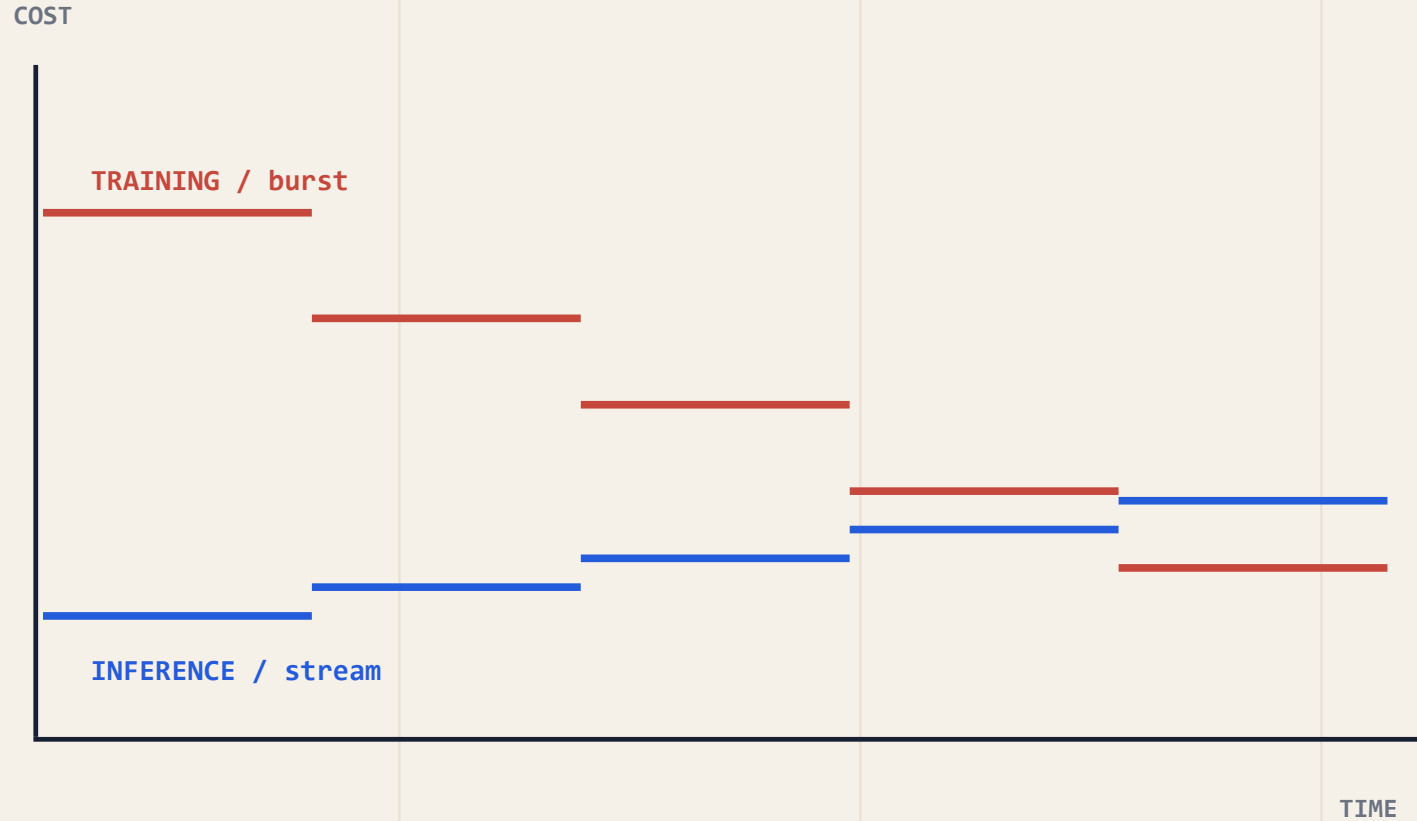
depreciation + margin

OPERATING
LEVERAGE

NO UTILIZATION → NO LEVERAGE

Training is episodic. Inference is continuous.

Illustrative curves show why utilization and serving efficiency can matter more than a single benchmark.



OPERATOR QUESTION

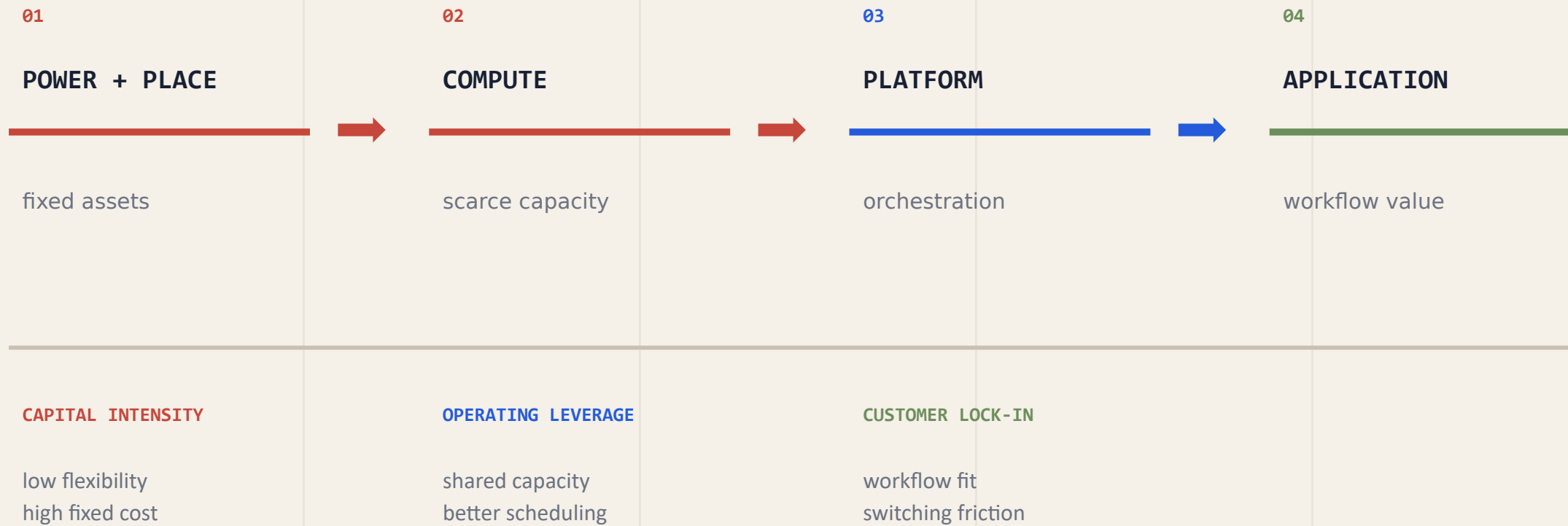
Can the load be scheduled, shifted, or pooled?

ILLUSTRATIVE ONLY

Shape of curve is conceptual not a benchmark

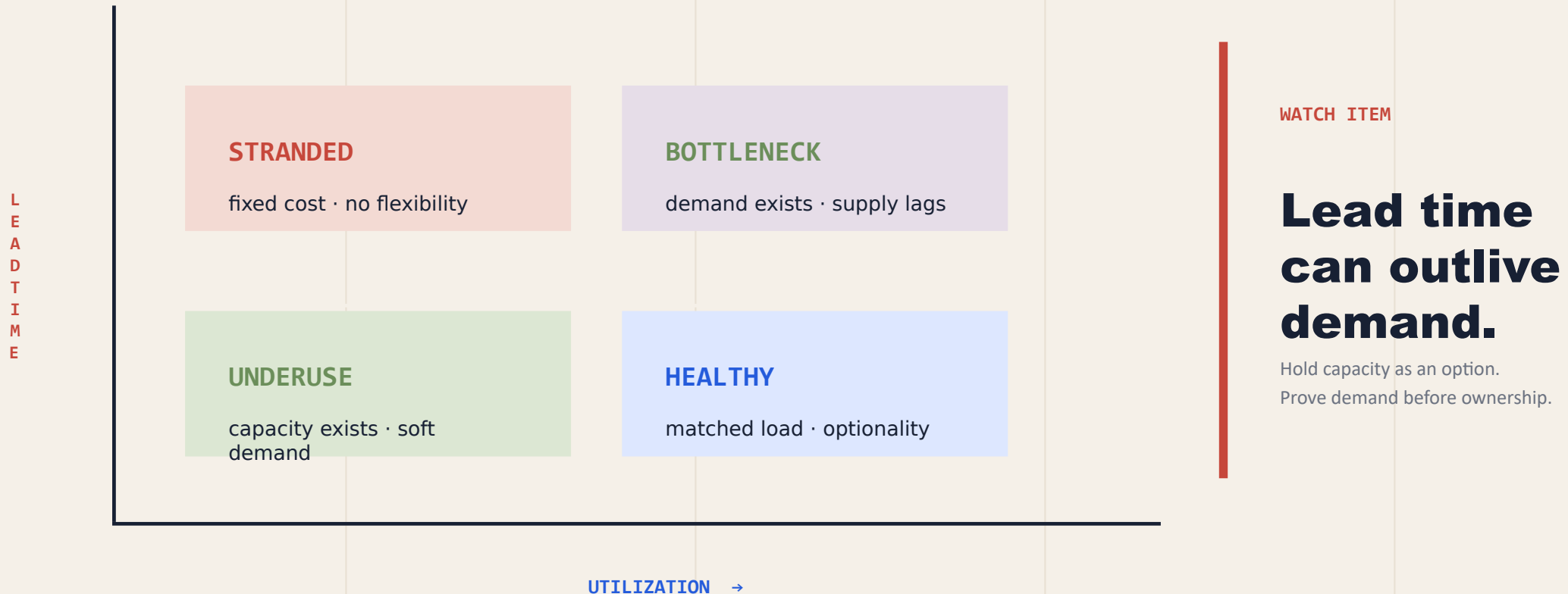
The margin pool is a flow, not a single layer.

Compute creates the possibility; platforms and applications decide where value compounds.



The dangerous asset is capacity you cannot use.

A simple matrix makes the hidden risk visible: high lead time plus low utilization.



There are three ways to buy capacity. None is free.

The right answer depends on demand certainty, capital appetite, and control requirements.

BUY

control

CAPITAL high

SPEED medium

CONTROL high

RISK demand

RENT

optionality

variable

fast

shared

vendor

BUILD

differentiation

very high

slow

highest

execution

OPERATOR TEST

How certain
is demand?
How unique
is the
workload?

Run infrastructure like an operating system, not a warehouse.

Five measures reveal whether the asset is earning its keep.

01

CAPACITY

what exists

02

UTILIZATION

what is used

03

LATENCY

what users feel

04

DEPRECIATION

what time costs

05

POWER

what scale requires

THE SCORECARD TURNS CAPEX INTO A FEEDBACK SYSTEM.

If one measure is missing, the operator is managing a blind spot — not a platform.

OBSERVE → OPERATE → REINVEST

The winning infrastructure is not the biggest. It is the most legible.

Capital creates the option. Constraints define the clock. Control decides the return.

Capital creates the option.

Constraints define the clock.

Control decides the return.

THE OPERATOR'S QUESTION

Where is the bottleneck?

What can we flex?

What must we own?

MAKE THE STACK LEGIBLE.